

Sociotechnical Risk in Sustaining Digital Infrastructure

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ABSTRACT

Significant risks to our shared digital infrastructure—communication systems, servers, and applications—can be identified by examining the participation dynamics of the organizations which produce that infrastructure. Exploration of these production communities reveals the deeply contingent processes of collective action that sustain them—processes and organizations that are innovative and powerful but sometimes fragile. As this shared body of digital infrastructure has grown, some crucial pieces have become neglected, leading to *underproduction*: the phenomenon of highly important, low-quality digital artifacts. This dissertation is framed around a series of empirical projects. The first will examine participation dynamics with respect to existing measures of underproduction. I then develop a longitudinal measure of underproduction to understand how underproduction emerges over time. Finally, I apply this measure to digital infrastructure organizations longitudinally, as participation dynamics are unfolding.

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1 PROBLEM DOMAIN: DIGITAL INFRASTRUCTURE

When I buy airline tickets, a little lock in the URL bar tells me my purchase is secure. When I forward an e-mail of my itinerary to my hosts, it arrives in seconds. When I step off the plane, my phone updates my timezone and synchronizes the clock. The flow of information that sustains these mundane miracles is the quiet evidence of digital infrastructure in modern life. Each of these innovations is deeply dependent on a multi-decade legacy of effort—often by volunteers worldwide—to create new organizational forms, to develop innovative technology, and to sustain the vast array of information public goods that make many communication practices possible. The maintenance of information public goods such as free/libre open source software often relies on commons-based peer

production [1] (CBPP). Developing and maintaining this software through CBPP involves novel organizational forms in which self-organizing volunteers define a digital artifact in terms that allow for modular efforts, select their own tasks, and contribute to a shared goal. Although the results of these efforts are often innovative and influential (e.g. the software powering the web and cloud), the tasks being done by self-selecting volunteers may or may not align with those tasks most deeply needed to sustain services to the general public.

Trying to see the infrastructure that underpins familiar platforms reveals a dizzying array of technologies: they are hosted in clouds that cross datacenters, composed of servers running containers running software that, in many cases, has developed and is maintained via cooperative, voluntary, and highly contingent efforts. In the midst of this maze of technology, we find a profoundly human process: joyful moments of problem solving, frustrated calls for help, labors of love crafted into greatness, and moments of kindness to strangers. The sprawling, interdependent, and fundamentally social organizations that build and maintain digital infrastructure are at times flawed, and their products include neglected but vital software packages. Some of these neglected packages are buried deep in the software supply chain and their failure may not be noticed until they cause disruptions—a condition which places modern society at risk. Of particular concern is *underproduced* software—highly important, low quality packages. How can we identify these risks and remediate them before they cause major failures? One angle of approach is to better understand the organizations producing and maintaining digital infrastructure and to then identify the organizational patterns associated with underproduction.

2 RESEARCH QUESTIONS

This dissertation will develop evidence in support of my thesis that we can identify underproduction risks to digital infrastructure by examining the state of the development organizations maintaining it.

I will answer three key research questions in pursuit of this thesis:

- (1) How are membership and participation dynamics associated with underproduction?
- (2) How can we measure underproduction within an organization?
- (3) How does underproduction emerge over time?

3 RELATED WORK

Collaborative communities engaged in a form of collective action called commons-based peer production have produced a wide range of information public goods [1, 2, 13, 16]. In commons-based peer production, individuals make modular contributions of varying

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sizes that can then be combined together in a relatively low-cost fashion. The end result of this distributed community-based effort is a body of information public goods upon which our society relies, including key components in the functioning of the web, e-mail, cell phones, and financial services [10]. One advantage of commons-based peer production as Benkler [2] pointed out is the efficiency gained by supporting self-directed contributions: each contributor is an expert in their own skills and interests, matching their capabilities, values, and creativity to a problem of their choosing. Further, Benkler observed that modularity reduces coordination costs while also allowing for flexible levels of commitment. The massively distributed character of software production decisions leads to a decentralized power structure that in turn creates space for experimentation in how people collaborate and in what is created.

The efficiencies and advantages of commons-based peer production are accompanied by certain inefficiencies and disadvantages. The interests of self-directed developers may not accord with what work is most needed by the public. Decentralization makes it difficult for developers, much less users, to know when this misalignment of effort is occurring, or even to keep track of the quality and importance of packages. Software development organizations may grow organically around a small group of high-volume contributors, then struggle to adapt their structure as needs and conditions change. Original assumptions about code design and software features may not age well or prove unsustainable. Nevertheless, commons-based peer production has over the last three decades produced innovations in the form of information public goods.

The study of how communities organize their membership—who joins, how they are onboarded and socialized, what roles members take up with respect to one another, and how they transition out of the organization—has long been an area of interest both for both Computer-Supported Cooperative Work scholars and for Communication scholars. Scholars in Computer-Supported Cooperative Work and Social Computing have examined peer production organizations to understand how members onboard and are socialized into organizational norms (e.g. [4]), the roles members take up (e.g. [14]), the unequal “core-periphery” structure that forms among participants [3, 8, 9], and concerns about developing sustainable structures [5]. Ultimately, as Lee [15] and others have pointed out, these organizations and their members form the “human infrastructure” that sustains digital infrastructure.

Scholars in Communication have likewise investigated peer production organizations, elaborating information public goods as both connective and communal [13] with fluid roles [12] but also displaying an unequal structure [19]. Communication scholars have further argued that all organizations are constituted by communication [17] as they engage with their shared goals. McPhee and Zaig (2000) offer the Four Flows model as a way to describe this constituting perspective, offering that organizations are defined and composed by four forms of communication: the ways in which they negotiate membership, self-structure, coordinate activity, and position their organization within a broader range of sociotechnical communities. My dissertation is focused on developing empirical evidence with respect to one of these communication flows: *membership negotiation*.

Although peer production has proven to be a remarkably successful way to create information public goods, these goods are subject to **social production failures**, a concept I propose as an extension and novel application of the concepts of market failures and social failures. A market failure occurs when a mutually beneficial exchange does not occur [7], while a social failure occurs when a mutually beneficial social interaction does not occur [18]. By extension, a social production failure can occur if some mutually beneficial social production activity fails or is not sustained. The present work is focused on a single form of social production failure: underproduction. Underproduction was originally elaborated in economics as a form of market failure—that is, a case where a mutually beneficial exchange does not occur—and refers to the case when demand far outstrips supply. In the case of information public goods, I define underproduction as the condition that arises when the quality of these goods is relatively low while their importance is relatively high. Underproduction in the case of digital infrastructure is of particular concern [10, 11]. My thesis is that we can identify and trace the underproduction of digital infrastructure by examining the structural communication dynamics of the organizations producing that infrastructure.

4 CONTRIBUTIONS

This work provides insight into important puzzles in CSCW: How do organizations engaged in complex collaboration and coordination succeed enough that their efforts gain broad importance, but then fail in ways that put us at risk? How can we support communities in creating information public goods that are not only innovative and empowering but also can be sustained?

Peer production in pursuit of information public goods is a highly effective form of computer-supported cooperative work which has shaped the modern world. Scholars in social computing and CSCW may find particular value in both the empirical findings and methods developed through this dissertation.

The body of work resulting from this dissertation will deliver on three key contributions:

- (1) a model of organizational sources of underproduction (*empirical*)
- (2) a within-package underproduction measure, suitable for longitudinal analysis (*methodological*)
- (3) a longitudinal analysis of underproduction using organizational traits (*empirical*)

5 WORKPLAN

My proposal was approved by my doctoral thesis committee on June 10, 2022. I plan to complete and defend the dissertation by June 2023. I am currently applying for funding and expanding my network of organizational partnerships to support this work.

The dissertation is organized around three projects which inform one another. Although the dissertation projects are building on pilot work and previous projects, what I am describing is largely new work that is yet to be done. I will be extracting data from software repositories as well as making use of archives created and maintained by software development organizations.

5.1 Ethics

This work makes use of public datasets and analytical techniques that have been determined by the University of Washington IRB to not be human subjects research. However, I acknowledge that my work is only possible because of the longstanding efforts of global communities to build and sustain digital infrastructure. Given the ways in which I have benefited from this collective effort, I believe that adopting open science practices and engaging with practitioner communities are my ethical obligation.

As I develop my work, I am committed to continuing to share this work broadly with the public and with stakeholder groups. Some of those key stakeholders are digital infrastructure development and maintenance communities (e.g. the Debian Project). This area of work has also received substantial interest from civil society organizations, and my intention is to continue to engage with them as well (e.g. Sloan Foundation, Ford Foundation, and the Linux Foundation, especially the CHAOSS project). This ethical commitment to dissemination takes the form of seeking out opportunities to present work in practitioner spaces, producing practitioner-friendly publications that summarize research in actionable ways, and a willingness to say “yes” when invited to engage with new communities.

5.2 Project 1: How are membership and participation dynamics associated with underproduction?

In the first of these three projects, I will take existing cross-sectional data and examine associated organizational failures. Previous research has explored many challenges that digital infrastructure organizations face with respect to membership and participation, including excessive reliance on single individuals and failure to engage contributors in systematic work processes. To what extent are these challenges associated with significant organizational failures like the neglect of important packages (underproduction)? In this project, I will use a cross-sectional approach drawn from my prior work [6] to test relationships between underproduction and two measures associated with membership and participation: maintainer tenure and centrality in social networks inside packages.

Hypothesis: Underproduction is associated with unstable or absent leadership, and excessively centralized collaboration structures

Methods: Underproduction analysis [6] and network analysis (nodes: people; edges: having both contributed to a bug-related discussion)

5.3 Project 2: How can we measure underproduction within an organization?

The second project will focus on developing better measures and methods. Measures of underproduction in my prior work [6] are oriented to relative underproduction at a macro level, e.g. at-risk packages inside a software distribution of thousands of packages, or neglected articles within a body of millions of articles. This allows assessment of historical overall neglect but does not reveal whether a given organization is currently in a state of risk. In this chapter, I will detail and demonstrate a method for measuring within-artifact

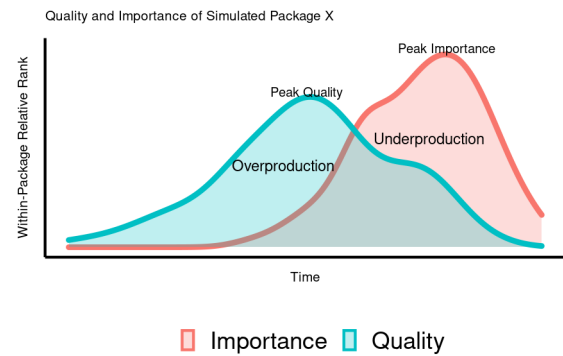


Figure 1: Software quality and software importance sometimes diverge. When quality exceeds importance (“underproduction”), our shared digital infrastructure is at risk.

underproduction using longitudinal data. This project will make use of digital trace data to analyze software *quality* and *importance*.

In a longitudinal framework, the key indicators for underproduction are the current gap between quality and importance, the points of greatest absolute distance between quality and importance, and the area between the quality and importance curves. In the simulated result depicted in Figure 1, we see that the importance of the package, in red, continues to climb even after the quality of the package, in blue, has entered a phase of decline. My claim is that services using package X, or dependent on others using it, are being exposed to substantial risk during the underproduction period.

Hypothesis: Software quality and importance rise and then fall through time.

Methods: Underproduction analysis [6], time series analysis, software quality measures, software importance measures

5.4 Project 3: How does underproduction emerge over time?

The third and final project, informed by the findings of the first project and using the methods of the second project, takes a longitudinal view. Studies of free/libre open software have often remarked on its core-periphery structure, in which a small number of people conduct an outside volume of the effort. This structure is often viewed as a source of risk; this chapter will take the perhaps contrarian position that participation inequality (e.g. oligarchy [19]) may offers some important benefits at some points in the lifecycle of a project and that the key concern for organizations should be to know under what conditions these benefits are likely to be outweighed by the risks.

Hypothesis: To avoid underproduction as a project shifts from early development into maturity, software development organizations must successfully negotiate a membership structure that allows for a core group with moderately broad participation.

Methods: Underproduction analysis [6] and contribution network analysis (e.g. nodes: contributors; edges: having contributed to the same file within a software project)

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A QUESTIONS FOR THE CONSORTIUM

- (1) *Maintaining a “Science” Focus* I am eager for this work to be used by others, but I also don’t want to lose track of the basic science focus: I’m not trying to build yet another measure or feed a dashboard. I want to reveal structural processes and collaboration dynamics. How can I keep both of these interests in balance?
- (2) *Contribution* Part of my intended contribution to CSCW is to provide a stronger bridge between Communication (and the subfield of Organizational Communication) and HCI. What are the most successful approaches for doing work that effectively bridges two very different academic traditions?
- (3) *Methods* My approach as proposed relies on network analysis methods, however I am comfortable with a range of quantitative and mixed-methods approaches. Are there methodological alternatives I should be considering for deriving measures of collaboration and inequality?

B BIOGRAPHICAL SKETCH

Kaylea Champion is a PhD Candidate in the Department of Communication at the University of Washington. Her anticipated dissertation completion date is June 2023. Prior to her work at the University of Washington, she was an IT director, consultant, and administrator. She holds a MS in Computer Science and an MA in Critical and Creative Thinking.

C CONNECTIONS AND INSPIRATIONS

Although my recent work in underproduction analysis [6] is directly connected to my dissertation, I find Elinor Ostrom’s work in *Governing the Commons* to be most broadly inspirational.

D FUNDING AVAILABILITY

I have access to some funding to support my participation and am currently pursuing additional support.